

Business Problem Statement

A leading apparel retail company wants to optimize its supply chain operations in order to reduce costs, improve product availability, and ensure timely delivery across all channels. The operations team has identified recurring inefficiencies across procurement, inventory management, and order fulfillment for its three core product lines including Pants, Jeans, and T-Shirts. They are particularly concerned about stockouts during peak demand seasons, unreliable supplier lead times, and the absence of real-time visibility into warehouse and distribution performance.

You are tasked with analyzing the company's supply chain dataset to answer the following overarching business question:

“How can the company use supply chain data to find problems, plan inventory better, and make the order-to-delivery process more efficient for Pants, Jeans, and T-Shirts?”

Deliverables

- 1. Data Preparation & Modeling (Python):** Clean and transform the raw supply chain dataset, handle missing values, fix inconsistent records, and engineer key features such as lead time categories, stock health flags, fulfillment status, and profit margin per product.
- 2. Data Analysis (SQL):** Organize the data into a structured relational schema, simulate supply chain transactions, and run queries to extract insights on inventory turnover, supplier performance, order cycle times, stockout frequency, and category-level demand trends.
- 3. Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key supply chain KPIs and trends — including inventory health, fulfillment rates, supplier scorecards, and seasonal demand patterns — enabling stakeholders to make data-driven operational decisions.
- 4. Report and Presentation:** Write a clear project report summarizing key findings and supply chain recommendations. Prepare a presentation that visually communicates insights and actionable strategies to management stakeholders.
- 5. GitHub Repository:** Include all Python scripts, SQL queries, and Power BI dashboard files in a well-structured repository with proper documentation and a README.